

CSE-214 Online - 2 (A1)

Structural Design Pattern

Duration: 30 minutes

A gift shop sells gift items such as showpieces, decorative ornaments, souvenirs, etc. Each gift item has a description and a base price. Customers may optionally request that a purchased gift be wrapped, which adds **\$2** to the item's price. Customers may also request home delivery. Initially, the shop provided delivery only within the local city, charging **\$1** per mile from the shop to the destination, with an estimated delivery time of 1 week. Later, the shop expanded its service to support national delivery, where the delivery charge is \$1 per mile plus a fixed surcharge of \$20, with an estimated delivery time of 1–2 weeks.

The shop is now planning to introduce international delivery for selected countries. International delivery incurs a fixed surcharge of **\$500** and has an estimated delivery time of 2–3 weeks. In addition, customers may choose one of the following delivery modes:

Express Delivery: Adds \$10 to the delivery charge. The estimated delivery time is 2 days for local and national deliveries, and 1 week for international deliveries.

Priority Delivery: Adds \$25 to the delivery charge. The estimated delivery time is 1 day for local and national deliveries, and 5 days for international deliveries.

The shop expects to introduce additional delivery regions and delivery modes in the future. The design should support extending both independently without requiring significant modifications to the existing implementation. Your task is to implement the whole system so that a product's price and delivery time (if applicable) can be calculated and shown properly. Choose one or multiple structural design patterns carefully. Some example cases are given below:

Case 1

A decorative vase costing **\$40** will be sent to a location **10 miles** away within the local city with gift wrapping.

Total Cost:

$$\begin{aligned} &= \text{Gift Price} + \text{Wrapping Charge} + \text{Local Delivery Charge} \\ &= \$40 + \$2 + (10 \times \$1) \\ &= \$52 \end{aligned}$$

Estimated Delivery Time: 1 week

Case 2

A wooden souvenir costing **\$60** will be sent to a location **50 miles** away within the country with **gift wrapping** and **Express Delivery**.

Total Cost:

$$\begin{aligned} &= \text{Gift Price} + \text{Wrapping Charge} + \text{National Delivery Charge} + \text{Express Charge} \\ &= \$60 + \$2 + (50 \times \$1 + \$20) + \$10 \\ &= \$142 \end{aligned}$$

Estimated Delivery Time: 2 days

Case 3

A crystal showpiece costing **\$150** will be shipped to an eligible international destination using **Priority Delivery**.

Total Cost:

$$\begin{aligned} &= \text{Gift Price} + \text{International Delivery Charge} + \text{Priority Charge} \\ &= \$150 + \$500 + \$25 \\ &= \$675 \end{aligned}$$

Estimated Delivery Time: 5 days